

Day 13 – Linux Volume Management (LVM)

LVM (Logical Volume Management) helps you manage storage dynamically in Linux.

You can:

- Create flexible partitions
 - Extend storage without reboot
 - Manage disks easily
-

Step 1: Create a Virtual Disk (If No Extra Disk Available)

Create a 1GB virtual disk file:

None

```
dd if=/dev/zero of=/tmp/disk1.img bs=1M count=1024
```

Meaning:

- `if=/dev/zero` → input zeros
- `of=` → output file
- `bs=1M` → block size 1MB
- `count=1024` → total 1024 MB = 1GB

Expected output:

None

```
1024+0 records in  
1024+0 records out
```

Step 2: Attach Virtual Disk as Loop Device

Run:

None

```
losetup -fP /tmp/disk1.img
```

Check attached loop device:

None

```
losetup -a
```

Example output:

None

```
/dev/loop0: []: (/tmp/disk1.img)
```

Your device may be:

- `/dev/loop0`
- `/dev/loop1`

Use YOUR device name in next commands.

Step 3: Check Current Storage

Run these commands one by one:

None

```
lsblk
```

Shows disks and partitions.

```

root@ip-172-31-17-115:/home/ubuntu# lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0  28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                 7:1      0   74M   1 loop /snap/core22/2411
loop2                 7:2      0  48.4M  1 loop /snap/snapd/26382
loop3                 7:3      0 454.4M  0 loop
nvme0n1              259:0     0    8G   0 disk
├─nvme0n1p1          259:1     0   6.9G  0 part /
├─nvme0n1p13         259:2     0 1023M  0 part /boot
├─nvme0n1p14         259:3     0    4M   0 part
└─nvme0n1p15         259:4     0  106M  0 part /boot/efi
root@ip-172-31-17-115:/home/ubuntu#

```

Difference between lsblk and df -h

```

root@ip-172-31-17-115:/home/ubuntu# lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0  28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                 7:1      0   74M   1 loop /snap/core22/2411
loop2                 7:2      0  48.4M  1 loop /snap/snapd/26382
loop3                 7:3      0 454.4M  0 loop
nvme0n1              259:0     0    8G   0 disk
├─nvme0n1p1          259:1     0   6.9G  0 part /
├─nvme0n1p13         259:2     0 1023M  0 part /boot
├─nvme0n1p14         259:3     0    4M   0 part
└─nvme0n1p15         259:4     0  106M  0 part /boot/efi
root@ip-172-31-17-115:/home/ubuntu# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G  31% /
tmpfs            455M   0  455M   0% /dev/shm
tmpfs            182M  888K  181M   1% /run
efivarfs         128K   3.3K  120K   3% /sys/firmware/efi/efivars
tmpfs            455M  455M    0 100% /tmp
none             1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
none             1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M   96M  826M  11% /boot
/dev/nvme0n1p15  105M   6.3M   99M   7% /boot/efi
none             1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
none             1.0M    0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            91M   8.0K   91M   1% /run/user/1000
root@ip-172-31-17-115:/home/ubuntu#

```

None

pvs

Shows physical volumes.

None

```
vgs
```

Shows volume groups.

None

```
lvs
```

Shows logical volumes.

None

```
df -h
```

Shows mounted filesystem storage.

Take screenshots here.

Step 4: Create Physical Volume (PV)

If using loop device:

None

```
pvccreate /dev/loop0
```

If using extra disk:

None

```
pvccreate /dev/sdb
```

Expected output:

None

Physical volume `"/dev/loop0"` successfully created.

Verify:

None

`pvs`

Step 5: Create Volume Group (VG)

Create VG named `devops-vg`:

None

`vgcreate devops-vg /dev/loop0`

OR

None

`vgcreate devops-vg /dev/sdb`

Expected output:

None

Volume group `"devops-vg"` successfully created

Check:

None

`vgs`

Step 6: Create Logical Volume (LV)

Create 500MB logical volume:

None

```
lvcreate -L 500M -n app-data devops-vg
```

Meaning:

- `-L 500M` → size
- `-n app-data` → LV name

Expected output:

None

```
Logical volume "app-data" created.
```

Check:

None

```
lvs
```

Step 7: Format the Logical Volume

Format using EXT4 filesystem:

None

```
mkfs.ext4 /dev/devops-vg/app-data
```

Expected output:

None

```
Creating filesystem with ...
```

Step 8: Create Mount Directory

None

```
mkdir -p /mnt/app-data
```

Step 9: Mount the Logical Volume

None

```
mount /dev/devops-vg/app-data /mnt/app-data
```

Verify:

None

```
df -h /mnt/app-data
```

Expected output example:

None

```
Filesystem                                Size Used Avail Use% Mounted on
/dev/mapper/devops--vg-app--data          500M ...
```

Step 10: Extend the Logical Volume

Increase storage by 200MB:

None

```
lvextend -L +200M /dev/devops-vg/app-data
```

Expected output:

None

```
Size of logical volume changed from 500 MiB to 700 MiB
```

Step 11: Resize Filesystem

Very important step:

None

```
resize2fs /dev/devops-vg/app-data
```

Without this, filesystem size won't increase.

Step 12: Verify Extended Storage

Run:

None

```
df -h /mnt/app-data
```

Now size should show around:

None

```
700M
```

Linux Volume Management (LVM)

Commands Used

None

```
sudo -i
dd if=/dev/zero of=/tmp/disk1.img bs=1M count=1024
losetup -fP /tmp/disk1.img
losetup -a
lsblk
pvs
vgs
lvs
df -h
pvcreate /dev/loop0
vgcreate devops-vg /dev/loop0
lvcreate -L 500M -n app-data devops-vg
mkfs.ext4 /dev/devops-vg/app-data
mkdir -p /mnt/app-data
mount /dev/devops-vg/app-data /mnt/app-data
df -h /mnt/app-data
lvextend -L +200M /dev/devops-vg/app-data
resize2fs /dev/devops-vg/app-data
df -h /mnt/app-data
```

Important DevOps Interview Concepts

LVM Hierarchy

None

Disk → PV → VG → LV → Filesystem → Mount Point

Example:

None

/dev/sdb

↓

Physical Volume

↓

Volume Group (devops-vg)

↓

Logical Volume (app-data)

↓

ext4 filesystem

↓

/mnt/app-data

Useful Commands for Troubleshooting

Check mounted volumes:

None

mount | grep app-data

Check detailed LV info:

None

lvdisplay

Check detailed VG info:

None

vgdisplay

Check detailed PV info:

None

pvdisplay

Real DevOps Usage

LVM is used in:

- AWS EC2 storage expansion
- Database servers
- Kubernetes worker nodes
- Application servers
- Dynamic disk management in production Linux servers

Mine: extra video-----> shubham

```
ubuntu@ip-172-31-45-118:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1       7:1    0 48.4M  1 loop /snap/snapd/26382
loop2       7:2    0  74M   1 loop /snap/core22/2411
nvme0n1     259:0    0   8G   0 disk
├─nvme0n1p1 259:1    0  6.9G   0 part /
├─nvme0n1p13 259:2    0 1023M   0 part /boot
├─nvme0n1p14 259:3    0    4M   0 part
└─nvme0n1p15 259:4    0  106M   0 part /boot/efi
ubuntu@ip-172-31-45-118:~$
```

nvme0n1-root-----> /dev/nvme0n1

```
ubuntu@ip-172-31-45-118:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       6.7G  2.1G  4.6G  31% /
tmpfs           455M    0 455M   0% /dev/shm
tmpfs           182M 888K 181M   1% /run
```

```

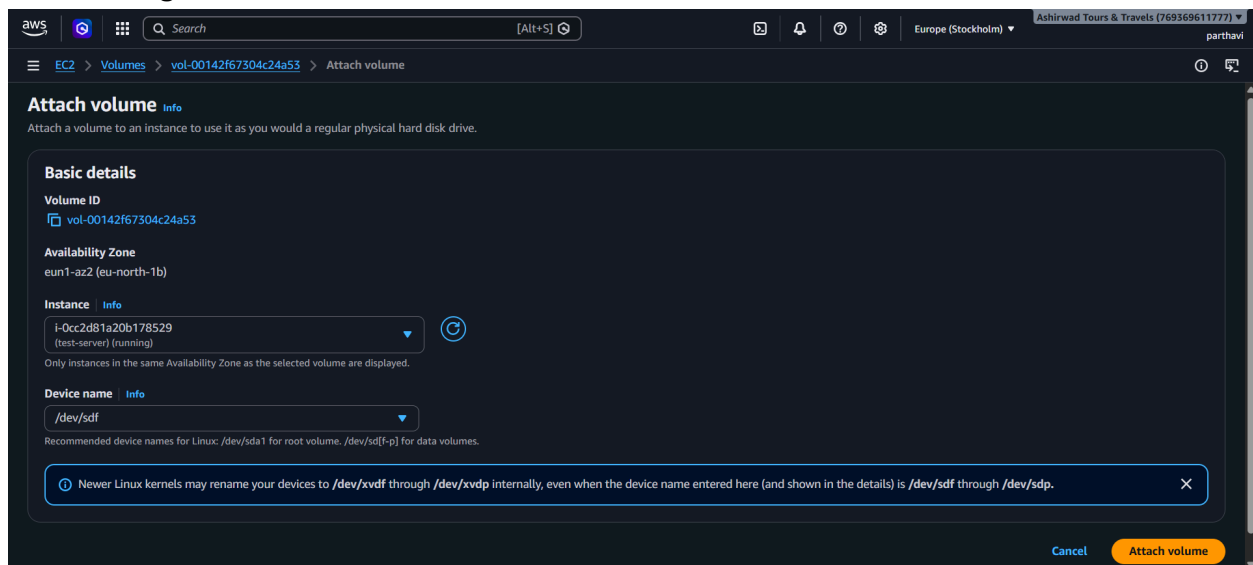
efivarfs      128K 3.3K 120K  3% /sys/firmware/efi/efivars
tmpfs         455M  0 455M  0% /tmp
none          1.0M  0 1.0M  0% /run/credentials/systemd-journald.service
none          1.0M  0 1.0M  0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 989M 96M 826M 11% /boot
/dev/nvme0n1p15 105M 6.3M 99M  7% /boot/efi
none          1.0M  0 1.0M  0% /run/credentials/systemd-networkd.service
none          1.0M  0 1.0M  0% /run/credentials/getty@tty1.service
none          1.0M  0 1.0M  0% /run/credentials/serial-getty@ttyS0.service
tmpfs         91M 8.0K 91M  1% /run/user/1000
ubuntu@ip-172-31-45-118:~$

```

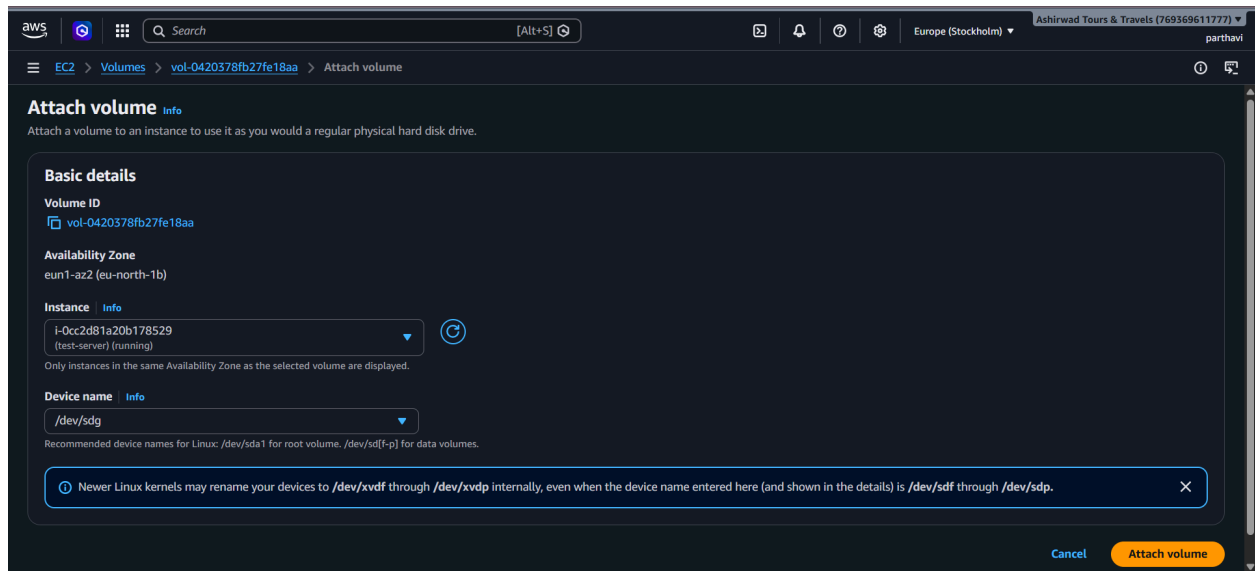
Ek location ke sath mount kar dena is mounting

Attach Volume:

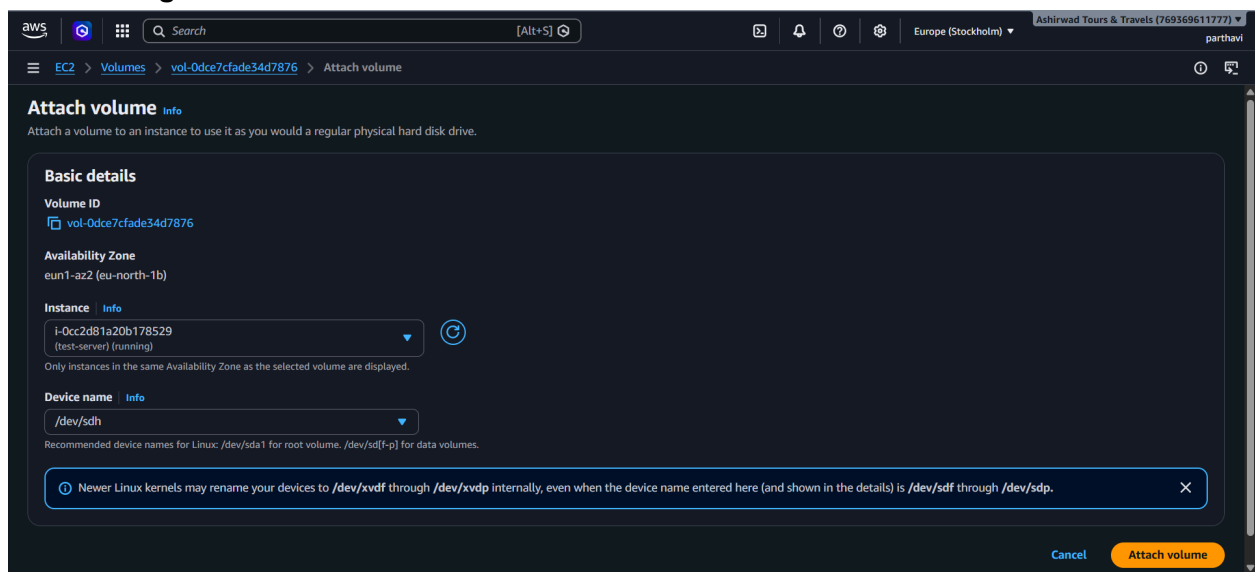
Start with f -----> /dev/sdf----->translate to /dev/xvdf in linux
Volume -10gb- /dev/sdf ----->/dev/



Volume-12gb- /dev/sdg----->/dev/



Volume- 15gb- /dev/sdh----->/dev/



```
root@ip-172-31-45-118:/home/ubuntu# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1       7:1    0 48.4M  1 loop /snap/snapd/26382
loop2       7:2    0  74M   1 loop /snap/core22/2411
nvme0n1     259:0    0   8G   0 disk
├─nvme0n1p1 259:1    0  6.9G   0 part /
├─nvme0n1p13 259:2    0 1023M   0 part /boot
├─nvme0n1p14 259:3    0    4M   0 part
└─nvme0n1p15 259:4    0  106M   0 part /boot/efi
```

nvme1n1	259:5	0	10G	0	disk
nvme2n1	259:6	0	12G	0	disk
nvme3n1	259:7	0	15G	0	disk

In Linux LVM (Logical Volume Management), these three terms are the foundation:

1. Physical Volume (PV)

A Physical Volume is the actual storage device.

Examples:

- Hard disk (`/dev/sdb`)
- SSD (`/dev/sdc`)
- Disk partition (`/dev/sdb1`)

Command:

None

```
pvcreate /dev/sdb
```

Think of it as:

 Raw storage space

2. Volume Group (VG)

A Volume Group is a pool of storage created by combining one or more Physical Volumes.

Example:

None

```
vgcreate myvg /dev/sdb /dev/sdc
```

Here:

- `/dev/sdb` = 100 GB
- `/dev/sdc` = 200 GB

VG size = 300 GB

Think of it as:

 Storage Pool

3. Logical Volume (LV)

A Logical Volume is the virtual partition created from the Volume Group.

Example:

None

```
lvcreate -L 50G -n mylv myvg
```

This creates a 50 GB logical volume from the 300 GB pool.

Think of it as:

 Virtual Partition

You can format and mount it:

None

```
mkfs.ext4 /dev/myvg/mylv  
mount /dev/myvg/mylv /data
```

Creating physical volume: `pvcreate /dev/xvdf /dev/xvdg /dev/xvdh`-----shubham

In my case: pvcreate /dev/nvme1n1 /dev/nvme2n1 /dev/nvme3n1

```
root@ip-172-31-45-118:/home/ubuntu# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                 7:1      0 48.4M  1 loop /snap/snapd/26382
loop2                 7:2      0   74M  1 loop /snap/core22/2411
nvme0n1               259:0     0    8G  0 disk
├─nvme0n1p1           259:1     0   6.9G  0 part /
├─nvme0n1p13          259:2     0 1023M  0 part /boot
├─nvme0n1p14          259:3     0    4M  0 part
└─nvme0n1p15          259:4     0  106M  0 part /boot/efi
nvme1n1               259:5     0   10G  0 disk
nvme2n1               259:6     0   12G  0 disk
nvme3n1               259:7     0   15G  0 disk
root@ip-172-31-45-118:/home/ubuntu# lvm
lvm> pvcreate /dev/nvme1n1 /dev/nvme2n1 /dev/nvme3n1
Physical volume "/dev/nvme1n1" successfully created.
Physical volume "/dev/nvme2n1" successfully created.
Physical volume "/dev/nvme3n1" successfully created.
lvm> █
```

pvs

```
root@ip-172-31-45-118:/home/ubuntu# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                 7:0      0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                 7:1      0 48.4M  1 loop /snap/snapd/26382
loop2                 7:2      0   74M  1 loop /snap/core22/2411
nvme0n1               259:0     0    8G  0 disk
├─nvme0n1p1           259:1     0   6.9G  0 part /
├─nvme0n1p13          259:2     0 1023M  0 part /boot
├─nvme0n1p14          259:3     0    4M  0 part
└─nvme0n1p15          259:4     0  106M  0 part /boot/efi
nvme1n1               259:5     0   10G  0 disk
nvme2n1               259:6     0   12G  0 disk
nvme3n1               259:7     0   15G  0 disk
root@ip-172-31-45-118:/home/ubuntu# lvm
lvm> pvcreate /dev/nvme1n1 /dev/nvme2n1 /dev/nvme3n1
Physical volume "/dev/nvme1n1" successfully created.
Physical volume "/dev/nvme2n1" successfully created.
Physical volume "/dev/nvme3n1" successfully created.
lvm> pvs
PV                VG Fmt  Attr PSize  PFree
/dev/nvme1n1      lvm2 ---  10.00g 10.00g
/dev/nvme2n1      lvm2 ---  12.00g 12.00g
/dev/nvme3n1      lvm2 ---  15.00g 15.00g
lvm> █
```

Physical volume created

Volume group create- vgcreate tws-vg /dev/nvme1n1 /dev/nvme2n1

```
lvm>
lvm> vgcreate tws-vg /dev/nvme1n1 /dev/nvme2n1
  Volume group "tws-vg" successfully created
lvm> █
```

Check- vgs

```
lvm>
lvm> vgcreate tws-vg /dev/nvme1n1 /dev/nvme2n1
  Volume group "tws-vg" successfully created
lvm> vgs
  VG      #PV #LV #SN Attr   VSize  VFree
  tws-vg   2   0   0 wz--n- 21.99g 21.99g
lvm> █
```

Create logical volume: lvcreate -L 10G -n tws_lv tws-vg

```
lvm>
lvm> vgcreate tws-vg /dev/nvme1n1 /dev/nvme2n1
  Volume group "tws-vg" successfully created
lvm> vgs
  VG      #PV #LV #SN Attr   VSize  VFree
  tws-vg   2   0   0 wz--n- 21.99g 21.99g
lvm> lvcreate -L 10G -n tws_lv tws-vg
  Logical volume "tws_lv" created.
lvm> █
```

Pvdisplay- shows all information

lvm> pvdisplay

--- Physical volume ---

PV Name	/dev/nvme1n1
VG Name	tws-vg
PV Size	10.00 GiB / not usable 4.00 MiB
Allocatable	yes
PE Size	4.00 MiB

Total PE 2559
Free PE 2559
Allocated PE 0
PV UUID F1azfM-S8Qs-3Xqv-egQm-BTOs-0r15-BsWred

--- Physical volume ---

PV Name /dev/nvme2n1
VG Name tws-vg
PV Size 12.00 GiB / not usable 4.00 MiB
Allocatable yes
PE Size 4.00 MiB
Total PE 3071
Free PE 511
Allocated PE 2560
PV UUID ItORTE-2roM-BxJj-Kanl-yfb4-yX6e-vivQwz

"/dev/nvme3n1" is a new physical volume of "15.00 GiB"

--- NEW Physical volume ---

PV Name /dev/nvme3n1
VG Name
PV Size 15.00 GiB
Allocatable NO
PE Size 0
Total PE 0
Free PE 0
Allocated PE 0
PV UUID XGtmJX-S1aq-BDtN-vO3Y-feVc-noJt-3lckVf

```

lvm> pvdisplay
--- Physical volume ---
PV Name               /dev/nvme1n1
VG Name               tws-vg
PV Size               10.00 GiB / not usable 4.00 MiB
Allocatable           yes
PE Size               4.00 MiB
Total PE              2559
Free PE               2559
Allocated PE          0
PV UUID               FlazfM-S8Qs-3Xqv-egQm-BT0s-0r15-BsWred

--- Physical volume ---
PV Name               /dev/nvme2n1
VG Name               tws-vg
PV Size               12.00 GiB / not usable 4.00 MiB
Allocatable           yes
PE Size               4.00 MiB
Total PE              3071
Free PE               511
Allocated PE          2560
PV UUID               ltORTe-2roM-BxJj-Kanl-yfb4-yX6e-vivQwz

"/dev/nvme3n1" is a new physical volume of "15.00 GiB"
--- NEW Physical volume ---
PV Name               /dev/nvme3n1
VG Name
PV Size               15.00 GiB
Allocatable           NO
PE Size               0
Total PE              0

```

Volume display- vgdisplay

```

lvm> vgdisplay
--- Volume group ---
VG Name               tws-vg
System ID
Format                lvm2
Metadata Areas        2
Metadata Sequence No  2
VG Access              read/write
VG Status              resizable
MAX LV                 0
Cur LV                1
Open LV                0
Max PV                 0
Cur PV                2

```

Act PV 2
VG Size 21.99 GiB
PE Size 4.00 MiB
Total PE 5630
Alloc PE / Size 2560 / 10.00 GiB
Free PE / Size 3070 / 11.99 GiB
VG UUID 6a8c1N-vZ39-eKpl-jFzl-T5nE-Hyoh-hgAbtm

```
lvm> vgdisplay
--- Volume group ---
VG Name                tws-vg
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   2
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 1
Open LV                 0
Max PV                  0
Cur PV                 2
Act PV                  2
VG Size                 21.99 GiB
PE Size                 4.00 MiB
Total PE                5630
Alloc PE / Size         2560 / 10.00 GiB
Free PE / Size          3070 / 11.99 GiB
VG UUID                 6a8c1N-vZ39-eKpl-jFzl-T5nE-Hyoh-hgAbtm
```

Logical display:

```
lvm> lvdisplay
--- Logical volume ---
LV Path                /dev/tws-vg/tws_lv
LV Name                 tws_lv
VG Name                 tws-vg
LV UUID                 l4cyo0-6VNy-haXw-PPgS-UU9d-Rnq9-slcDqi
LV Write Access         read/write
LV Creation host, time ip-172-31-45-118, 2026-05-31 12:45:19 +0000
LV Status                available
# open                   0
LV Size                  10.00 GiB
Current LE               2560
Segments                 1
Allocation                inherit
```

Read ahead sectors auto
- currently set to 256
Block device 252:0

lvm>

```
lvm> lvdisplay
--- Logical volume ---
LV Path                /dev/tws-vg/tws_lv
LV Name                 tws_lv
VG Name                 tws-vg
LV UUID                 I4cyo0-6VNY-haXw-PPgS-UU9d-Rnq9-sIcDqi
LV Write Access         read/write
LV Creation host, time ip-172-31-45-118, 2026-05-31 12:45:19 +0000
LV Status                available
# open                  0
LV Size                 10.00 GiB
Current LE              2560
Segments                1
Allocation               inherit
Read ahead sectors      auto
 - currently set to     256
Block device            252:0
```

Now lsblk

ubuntu@ip-172-31-45-118:~\$ lsblk

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
loop0	7:0	0	28.2M	1	loop	/snap/amazon-ssm-agent/13009
loop1	7:1	0	48.4M	1	loop	/snap/snapd/26382
loop2	7:2	0	74M	1	loop	/snap/core22/2411
nvme0n1	259:0	0	8G	0	disk	
└─nvme0n1p1	259:1	0	6.9G	0	part	/
└─nvme0n1p13	259:2	0	1023M	0	part	/boot
└─nvme0n1p14	259:3	0	4M	0	part	
└─nvme0n1p15	259:4	0	106M	0	part	/boot/efi
nvme1n1	259:5	0	10G	0	disk	
nvme2n1	259:6	0	12G	0	disk	
└─tws--vg-tws_lv	252:0	0	10G	0	lvm	
nvme3n1	259:7	0	15G	0	disk	

Mounting

1. mkdir /mnt/tws_lv_mount

```
root@ip-172-31-45-118:/home/ubuntu# mkdir /mnt/tws_lv_mount
root@ip-172-31-45-118:/home/ubuntu#
```

2. Format

mkfs.ext4 /dev/tws-vg/tws_lv

```
root@ip-172-31-45-118:/home/ubuntu# mkdir /mnt/tws_lv_mount
root@ip-172-31-45-118:/home/ubuntu# mkfs.ext4 /dev/tws-vg/tws_lv
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: f6f53816-fdeb-42c1-92bf-efb60ea71bcf
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-45-118:/home/ubuntu#
```

3. Mount

mount /dev/tws-vg/tws_lv /mnt/tws_lv_mount

```
root@ip-172-31-45-118:/home/ubuntu# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G  32% /
tmpfs            455M   0  455M   0% /dev/shm
tmpfs            182M  928K  181M   1% /run
efivarfs         128K   3.3K  120K   3% /sys/firmware/efi/efivars
tmpfs            455M   0  455M   0% /tmp
none             1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
none             1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M   96M  826M  11% /boot
/dev/nvme0n1p15  105M   6.3M   99M   7% /boot/efi
none             1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
none             1.0M   0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            91M   8.0K   91M   1% /run/user/1000
/dev/mapper/tws--vg-tws_lv 9.8G  2.1M  9.3G   1% /mnt/tws_lv_mount
root@ip-172-31-45-118:/home/ubuntu#
```

4. You can use now

```

root@ip-172-31-45-118:/home/ubuntu# cd /mnt/tws_lv_mount/
root@ip-172-31-45-118:/mnt/tws_lv_mount# mkdir devops
root@ip-172-31-45-118:/mnt/tws_lv_mount# █

```

Concept

Attach----- just block attached

Mount-----usable

5. Umount-----umount /mnt/tws_lv_mount

```

root@ip-172-31-45-118:~# umount /mnt/tws_lv_mount
root@ip-172-31-45-118:~# cd /mnt/tws_lv_mount/
root@ip-172-31-45-118:/mnt/tws_lv_mount# cd devops
bash: cd: devops: No such file or directory
root@ip-172-31-45-118:/mnt/tws_lv_mount# mount /dev/tws-vg/tws_lv /mnt/tws_lv_mount
root@ip-172-31-45-118:/mnt/tws_lv_mount# cd devops
root@ip-172-31-45-118:/mnt/tws_lv_mount/devops# █

```

Now using 15g-----ye disk hai

1. root@ip-172-31-45-118:~# mkdir /mnt/tws_disk_mount

```

root@ip-172-31-45-118:~# mkdir /mnt/tws_disk_mount
root@ip-172-31-45-118:~# cd /mnt
root@ip-172-31-45-118:/mnt# ls
tws_disk_mount  tws_lv_mount
root@ip-172-31-45-118:/mnt# █

```

2.format

```

root@ip-172-31-45-118:~# mkdir /mnt/tws_disk_mount
root@ip-172-31-45-118:~# cd /mnt
root@ip-172-31-45-118:/mnt# ls
tws_disk_mount  tws_lv_mount
root@ip-172-31-45-118:/mnt# mkfs -t ext4 /dev/nvme3n1
mke2fs 1.47.2 (1-Jan-2025)
/dev/nvme3n1 contains a LVM2_member file system
Proceed anyway? (y,N) y
Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: 3c97a364-fc89-4025-a372-8f230d28899c
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-45-118:/mnt# █

```

3. Mount and check df -h

```

root@ip-172-31-45-118:/mnt# mount /dev/nvme3n1 /mnt/tws_disk_mount/
root@ip-172-31-45-118:/mnt# df -h

```

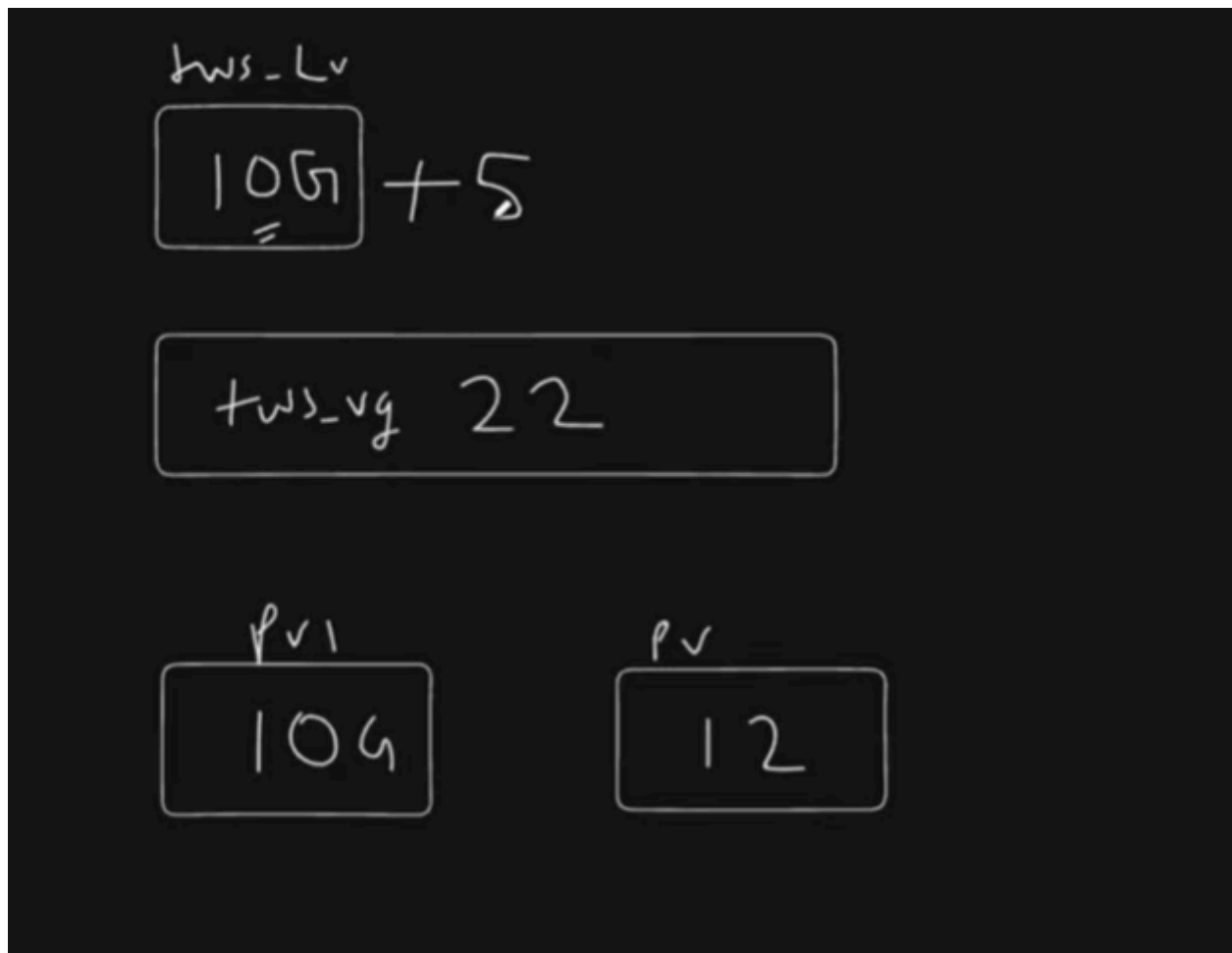
Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	6.7G	2.1G	4.6G	32%	/
tmpfs	455M	0	455M	0%	/dev/shm
tmpfs	182M	928K	181M	1%	/run
efivarfs	128K	3.3K	120K	3%	/sys/firmware/efi/efivars
tmpfs	455M	0	455M	0%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
/dev/nvme0n1p13	989M	96M	826M	11%	/boot
/dev/nvme0n1p15	105M	6.3M	99M	7%	/boot/efi
none	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
none	1.0M	0	1.0M	0%	/run/credentials/serial-getty@ttyS0.service
tmpfs	91M	8.0K	91M	1%	/run/user/1000
/dev/mapper/tws--vg-tws_lv	9.8G	2.1M	9.3G	1%	/mnt/tws_lv_mount
/dev/nvme3n1	15G	2.1M	14G	1%	/mnt/tws_disk_mount

```

root@ip-172-31-45-118:/mnt#

```

Extend extra 5gb



```
root@ip-172-31-45-118:/mnt# lvm
lvm> lvextend -L +5G /dev/tws-vg/tws_lv
  Size of logical volume tws-vg/tws_lv changed from 10.00 GiB (2560 extents) to 15.00 GiB (3840 extents).
  Logical volume tws-vg/tws_lv successfully resized.
lvm> █
```

lsblk

```
root@ip-172-31-45-118:/mnt# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                              7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                              7:1    0 48.4M  1 loop /snap/snapd/26382
loop2                              7:2    0  74M   1 loop /snap/core22/2411
nvme0n1                            259:0    0   8G   0 disk
├─nvme0n1p1                        259:1    0  6.9G  0 part /
├─nvme0n1p13                       259:2    0 1023M  0 part /boot
├─nvme0n1p14                       259:3    0    4M   0 part
└─nvme0n1p15                       259:4    0  106M  0 part /boot/efi
nvme1n1                            259:5    0  10G   0 disk
└─tws--vg-tws_lv                 252:0    0  15G   0 lvm  /mnt/tws_lv_mount
nvme2n1                            259:6    0  12G   0 disk
└─tws--vg-tws_lv                 252:0    0  15G   0 lvm  /mnt/tws_lv_mount
nvme3n1                            259:7    0  15G   0 disk /mnt/tws_disk_mount
root@ip-172-31-45-118:/mnt# █
```